CMSC 4200 - Artificial Intelligence
Spring 2026
Group Assignment 3

Due – Friday, March 6, 2026, at 11:59pm EST

Group #	1
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This assignment will be graded out of 30 points. Submit your answer document and program to the right D2L Dropbox named "Assignment 3 Dropbox" by the due date and time.

Question 1. (12 points) You are responsible for scheduling computer science classes that meet on Tuesday and Thursday. There are 8 classes that meet on these days and 4 professors who will be teaching these classes in which each professor will teach two classes. The classes are CMSC1200, CMSC1240, CMSC2040, CMSC3040, CMSC3100, CMSC3140, CMSC4000, CMSC4200. Classes meeting times are during 4 time slots: TU8:00-9:15AM, TU9:30-10:45AM, TU12:30-1:45PM, TU2:00-3:15PM.

The professors are:

- Professor 1 who is qualified to teach classes CMSC1200, CMSC3040, CMSC4000, and CMSC4200. He cannot teach at 8:00-9:15am.
- Professor 2 who is qualified to teach classes CMSC1200, CMSC1240, CMSC3040, CMSC3140, and CMSC4000. He cannot teach at 2:00-3:15pm.
- Professor 3 who is qualified to teach classes CMSC1200 and CMSC3100.
- Professor 4 who is qualified to teach classes CMSC1240, CMSC2040, and CMSC4200.

Formulate this problem as a constraint satisfaction problem (CSP). You need to clearly define the variables, the domains, and the constraints of this problem.

Information Known: 8 classes, 4 professors, each prof needs to teach 2 classes

Method: start with variables with lowest constraints

Time Slots
Tue/Th
TU8:00-9:15AM
TU9:30-10:45AM
TU12:30-1:45PM
TU2:00-3:15PM

Professor	Class	Can't Teach Time
Professor 1	CMSC1200,CMSC3040, CMSC4000, CMSC4200	8:00-9:15am.
Professor 2	CMSC1200,CMSC1240, CMSC3040,CMSC3140,CMSC4000.	2:00-3:15pm
Professor 3	CMSC1200 and CMSC3100	-no constraint
Professor 4	CMSC1240,CMSC2040,CMSC4200.	-no constraint

Shared:	
Professor	Class
Prof 1, Prof2, Prof 3	CMSC1200
Prof1, Prof2	CMSC3040, CMSC4000
Prof2, Prof4	CMSC1240
Prof1, Prof 4	CMSC4200

Possible Schedule:				
Professor:	Time 1:	CMSC Class 1:	Time 2:	CMSC Class 2:
Prof1	9:30-10:45AM	3040	2:00-3:15pm	4000
Prof2	8:00-9:15am.	3140	9:30-10:45AM	1240
Prof3	9:30-10:45AM	3100	12:30-1:45PM	1200
Prof4	12:30-1:45PM	2040	2:00-3:15pm	4200

***Problem formation**:*

X ={Professor1, Professor2, Professor3, Professor4 }

D_{time} = { TU8:00-9:15AM, TU9:30-10:45AM, TU12:30-1:45PM, TU2:00-3:15PM }

$D_{CMSC\ class}$ = {1200, 1240, 2040, 3040, 3100, 3140, 4000, 4200}

C_{time}= { Professor1 ≠ 8:00-9:15am, Professor2 ≠ 2:00-3:15pm}

$C_{class\ qualification}$ = {(Professor 1 = CMSC1200, CMSC3040, CMSC4000, and CMSC4200),
(Professor 2 = CMSC1200, CMSC1240, CMSC3040, CMSC3140, CMSC4000),
(Professor 3 = CMSC1200, CMSC3100),
(Professor 4 = CMSC1240, CMSC2040, CMSC4200)
}

***Goal**:* The goal of this CSP problem is to appropriately schedule each professor to teach 2 classes. They can't be scheduled to teach the same classes. The proposed schedule solution must also not conflict with their defined time constraints. Since a professor teaches two classes, their own classes must not conflict. No constraint is given on classrooms.

***Note**: Solution defined as *{Professor, (time1, class1) (time2, class2)}*

One Solution:

CSP = { Professor 1: (9:30-10:45am, 3040) (2:00-3:15pm, 4000),
Professor 2: (8:00-9:15am, 3140) (9:30-10:45am, 1240),
Professor 3: (9:30-10:45am, 3100) (12:30-1:45pm,1200),
Professor 4: (12:30-1:45pm, 2040) (2:00-3:15pm,4200)
}

Question 2. (18 points) Write a MiniZinc program to find a solution for Question 1. MiniZinc is a free and open-source constraint modeling language. For this question, you first need to watch the video - Tutorial Introduction to Minizinc on D2L. You can also watch other videos I posted with the assignment to help you write your MiniZinc program. You need to submit your MinZinc program and its output.

MiniZinc code (.mzn file submitted separately as well):

```
% You are responsible for scheduling computer science classes that meet on Tuesday and Thursday.

% There are 8 classes that meet on these days and 4 professors who will be teaching these classes in which each professor will teach two classes.

% The classes are CMSC1200, CMSC1240, CMSC2040, CMSC3040, CMSC3100, CMSC3140, CMSC4000, CMSC4200.

% Classes meeting times are during 4 time slots:  TU8:00-9:15AM, TU9:30-10:45AM, TU12:30-1:45PM, TU2:00-3:15PM.

% The professors are:

%       Professor 1 who is qualified to teach classes CMSC1200, CMSC3040, CMSC4000, and CMSC4200. He cannot teach at 8:00-9:15am.

%       Professor 2 who is qualified to teach classes CMSC1200, CMSC1240, CMSC3040, CMSC3140, and CMSC4000. He cannot teach at 2:00-3:15pm.

%       Professor 3 who is qualified to teach classes CMSC1200 and CMSC3100.

%       Professor 4 who is qualified to teach classes CMSC1240, CMSC2040, and CMSC4200.


%   Possible Classes, Times, and Professors

enum CLASS = {CMSC1200, CMSC1240, CMSC2040, CMSC3040, CMSC3100, CMSC3140, CMSC4000, CMSC4200};

enum TIME = {t_8to915, t_930to1045, t_1230to145, t_2to315};

enum PROFESSOR = {P1, P2, P3, P4};


%   Each class will get a time

array[CLASS] of var TIME: class_time;


%   Each class will get a professor

array[CLASS] of var PROFESSOR: teaches;
```

```
%   Each professor teaches 2 classes

constraint forall(p in PROFESSOR)(

    count(teaches, p) == 2

);


%   No professor teaches two classes at same time

constraint forall(c1, c2 in CLASS where c1 < c2)(

  (teaches[c1] == teaches[c2]) ->

  (class_time[c1] != class_time[c2])

);


%       Professor 1 who is qualified to teach classes CMSC1200, CMSC3040, CMSC4000, and CMSC4200. He cannot
teach at 8:00-9:15am.

constraint teaches[CMSC1240] != P1;

constraint teaches[CMSC2040] != P1;

constraint teaches[CMSC3100] != P1;

constraint teaches[CMSC3140] != P1;

constraint forall(c in CLASS)(

    (teaches[c] == P1) -> (class_time[c] != t_8to915)

);


%       Professor 2 who is qualified to teach classes CMSC1200, CMSC1240, CMSC3040, CMSC3140, and CMSC4000.
He cannot teach at 2:00-3:15pm.

constraint teaches[CMSC2040] != P2;

constraint teaches[CMSC3100] != P2;

constraint teaches[CMSC4200] != P2;

constraint forall(c in CLASS)(

    (teaches[c] == P2) -> (class_time[c] != t_2to315)

);


%       Professor 3 who is qualified to teach classes CMSC1200 and CMSC3100.

constraint teaches[CMSC1240] != P3;

constraint teaches[CMSC2040] != P3;

constraint teaches[CMSC3040] != P3;

constraint teaches[CMSC3140] != P3;

constraint teaches[CMSC4000] != P3;

constraint teaches[CMSC4200] != P3;
```

```
%       Professor 4 who is qualified to teach classes CMSC1240, CMSC2040, and CMSC4200.

constraint teaches[CMSC1200] != P4;

constraint teaches[CMSC3040] != P4;

constraint teaches[CMSC3100] != P4;

constraint teaches[CMSC3140] != P4;

constraint teaches[CMSC4000] != P4;


%   Solve the problem

solve satisfy;


%   Map times to more readble strings

array[TIME] of string: time_names = [

  "8:00-9:15AM",

  "9:30-10:45AM",

  "12:30-1:45PM",

  "2:00-3:15PM"

];


%   Output

output [

  "Professor 1 Schedule: " ++ show([ (c, time_names[fix(class_time[c])]) | c in CLASS where fix(teaches[c])
== P1 ]) ++ "\n" ++

  "Professor 2 Schedule: " ++ show([ (c, time_names[fix(class_time[c])]) | c in CLASS where fix(teaches[c])
== P2 ]) ++ "\n" ++

  "Professor 3 Schedule: " ++ show([ (c, time_names[fix(class_time[c])]) | c in CLASS where fix(teaches[c])
== P3 ]) ++ "\n" ++

  "Professor 4 Schedule: " ++ show([ (c, time_names[fix(class_time[c])]) | c in CLASS where fix(teaches[c])
== P4 ])

];
```

Output:

```
Assignment3 — powershell.exe
PS C:\Users\evant\Documents\CMSC-4200-AI\Assignment3> minizinc question2.mzn
Professor 1 Schedule: [(CMSC3040, "12:30-1:45PM"), (CMSC4000, "9:30-10:45AM")]
Professor 2 Schedule: [(CMSC1240, "8:00-9:15AM"), (CMSC3140, "9:30-10:45AM")]
Professor 3 Schedule: [(CMSC1200, "8:00-9:15AM"), (CMSC3100, "9:30-10:45AM")]
Professor 4 Schedule: [(CMSC2040, "9:30-10:45AM"), (CMSC4200, "8:00-9:15AM")]
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PS C:\Users\evant\Documents\CMSC-4200-AI\Assignment3>
```